

Classwork 06//KL Divergence

Student Name: _____ **Points:** 30

A Number: _____ **Due Date:** March 19, 2026

Consider a random variable $X \sim \text{Lognormal}(\mu, \sigma^2)$ with the probability density function (pdf):

$$p(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0$$

and normal distribution $Y \sim \mathcal{N}(m, v^2)$ with pdf

$$q(x) = \frac{1}{v\sqrt{2\pi}} \exp\left(-\frac{(x - m)^2}{2v^2}\right), \quad x \in \mathbb{R}$$

The KL divergence formula is

$$KL(p||q) = \int_0^\infty p(x) \log \frac{p(x)}{q(x)} dx$$

which is defined only if the support of P is contained in the support of Q . This is very important. Noting the above, compute the KL divergence $KL(\text{Lognormal}(\mu, \sigma^2)||\mathcal{N}(m, v^2))$.

Proof:

$$\log p(x) = -\log x - \log \sigma - \frac{1}{2} \log(2\pi) - \frac{(\ln x - \mu)^2}{2\sigma^2}$$

$$\log q(x) = -\log v - \frac{1}{2} \log(2\pi) - \frac{(x - m)^2}{2v^2}$$

$$\log \frac{p(x)}{q(x)} = -\log x - \log \sigma + \log v - \frac{(\ln x - \mu)^2}{2\sigma^2} + \frac{(x - m)^2}{2v^2}$$

$$KL(p||q) = \int_0^\infty p(x) \left[-\log x - \log \sigma + \log v - \frac{(\ln x - \mu)^2}{2\sigma^2} + \frac{(x - m)^2}{2v^2} \right] dx$$

$$KL(p||q) = -\mathbb{E}_p[\log x] - \log \sigma + \log v - \frac{1}{2\sigma^2} \mathbb{E}_p[(\ln x - \mu)^2] + \frac{1}{2v^2} \mathbb{E}_p[(x - m)^2]$$

From the property of lognormal distribution:
(log = ln)

$$\begin{aligned}\mathbb{E}[\ln x] &= \mu \\ \mathbb{E}[(\ln x - \mu)^2] &= \sigma^2\end{aligned}$$

And we apply the the proprty of expectations:

$$\begin{aligned}\mathbb{E}[(x - m)^2] &= \text{Var}(x) + (\mathbb{E}[x] - m)^2 = \mathbb{E}[x^2] + m^2 - 2m\mathbb{E}[x] \\ &= e^{2\mu+2\sigma^2} - 2me^{\mu+\frac{\sigma^2}{2}} + m^2\end{aligned}$$

Hence,

$$KL(p||q) = -\mu + \log \frac{v}{\sigma} - \frac{1}{2} + \frac{1}{2v^2}(e^{2\mu+2\sigma^2} - 2me^{\mu+\frac{\sigma^2}{2}} + m^2)$$